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Oil-price uncertainty and the U.K. unemployment rate: A forecasting experiment with random forests using 150 years of data☆

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ABSTRACT

We analyze the predictive role of oil-price uncertainty for changes in the UK unemployment rate using more than a century of monthly data covering the period from 1859 (when the drilling of the first oil well started at Titusville, Pennsylvania, United States) to 2020. To this end, we use a machine-learning technique known as random forests. Random forests render it possible to model the potentially nonlinear link between oil-price uncertainty and subsequent changes in the unemployment rate in an entirely data-driven way, where it is possible to control for the impact of several other macroeconomic variables and other macroeconomic and financial uncertainties. We estimate random forests on rolling-estimation windows and find evidence that oil-price uncertainty predicts out-of-sample changes in the unemployment rate, especially at longer (six and twelve months) forecast horizons. Moreover, the relative importance of oil-price uncertainty has undergone substantial swings during the history of the modern petroleum industry. Relative importance was high in the 1970s and the 1980s, and it was higher than the relative importance of changes in the oil price itself for most of the sample period. We also find that oil-price uncertainty has predictive value for changes in the unemployment rate when we use a Lasso estimator, where random forests have a superior forecasting performance relative to the Lasso forecasts.

1. Introduction

The (adverse) impact of oil-price volatility (uncertainty) on various macroeconomic aggregates (such as consumption and/or savings, investment, output, employment and/or unemployment, and international trade) has received considerable attention after the 1973 and 1979 oil-price shocks (following the Yom Kippur war and the Iranian Revolution), and continues to be so in the wake of the outbreak of the global pandemic of COVID-19 since the beginning of 2020. Intuitively, the negative effect of oil-price uncertainty (volatility) on real macroeconomic variables is generally explained by the real option theory (see for example, Bernanke (1983), Pindyck (1991), Dixit (1992), Dixit and Pindyck (1994), and more recently, Bloom (2009)), which suggests that decision making is affected by (oil-price) uncertainty because it raises the option value of waiting. In other words, given that the cost associated with wrong investment decisions is very high due to irreversibility, (oil-price) uncertainty makes firms and, in the case of durable goods, also consumers more cautious. As a result, economic agents postpone investment, hiring, and consumption decisions to periods of lower (oil-price) uncertainty, which results in cyclical fluctuations in macroeconomic aggregates. Given this theoretical channel suggesting a decline in various macroeconomic measures following a hike in oil-price uncertainty, a large international empirical literature has evolved trying to validate these claims (see for example, Guo and Kliesen (2005), Elder and Serletis (2009, 2010, 2011), Rafiq et al. (2009), Rahman and Serletis (2010, 2011, 2012), Bredin et al. (2011), Kilian and Vigfusson (2011), Bashar et al. (2013), Chen and Hsu (2013), Pinno and Serletis (2013), Aye et al. (2014), Jo (2014), Dave and Aye (2015), Elder (2018), van Eyden et al. (2019, 2021), Kandemir Kocaaslan (2019), Kocaarslan et al. (2020), Salisu et al. (2021), and the references cited therein, for earlier studies on this topic). In general,

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these studies find evidence in favor of the negative impact of oil-price uncertainty on consumption and/or savings, investment, output (i.e., real Gross Domestic Product (GDP), and industrial production), employment and/or unemployment, and even international trade.

These studies are indeed insightful, but as has been pointed out by Campbell (2008), the ultimate test of any predictive model (in terms of econometric methodologies and the predictors being used) is in its out-of-sample performance. Because existence of in-sample impact (predictability) does not necessarily ensure out-of-sample forecasting gains (Rapach and Zhou, 2013), our paper aims to provide a robust extension of the literature on oil-price uncertainty and its impact on an important macroeconomic variable by conducting an out-of-sample forecasting analysis of the predictive value of oil-price volatility for changes in the unemployment rate of the United Kingdom (UK) over the historical monthly period from 1859:10 to 2020:05. It should be noted that we focus on the forecastability of changes in the unemployment rate (rather than a metric of output) due to its availability over this long data span, besides being a prominent variable in capturing the position of the macroeconomy. In this regard, it should also be noted that the behavior of the unemployment rate can be tied with movements in output via the Okun's Law (Okun, 1970). The logic behind Okun's law is simple, and can be elaborated as follows: Output depends on the amount of labor used in the production process, so there is a positive relationship between output and employment. Further, total employment equals the labor force minus the unemployed, so there is a negative relationship between output and unemployment, or changes in the unemployment rate and the growth rate of output (as captured by real GDP and/or industrial production).

While our focus is oil-price uncertainty, to prevent omitted-variable bias, we incorporate a host of other macroeconomic and financial uncertainties (as well as the first-moments of these variables), the importance of which in forecasting macroeconomic aggregates has gained tremendous prominence in the wake of the "Great Recession" and the Global Financial Crisis (GFC) that followed thereafter (see for example, Karnizova and Li (2014), Balcilar et al. (2016), Junttila and Vataja (2018), Segnon et al. (2018), Aye et al. (2019a,b), Pierdzioch and Gupta (2020), Salisu et al. (2022)). Note that the choice of the UK as our case study is driven by the availability of a long span of data involving economic activity (as captured by the unemployment rate in our case) and a wide-array of predictors. Besides the data-availability issue, the UK is an important player in the oil market, with a standing of the 10th and 20th as an oil importer and exporter, respectively (Central Intelligence Agency (CIA) World Factbook, 2019). Hence, evidence of a predictive role of oil-price uncertainty for the development of the real economy, as captured by the unemployment rate would be a matter of valid concern for policymakers. It is important to understand that the need to look at a long historical period, which essentially covers the entire history of the modern petroleum industry that started with the drilling of the first oil well at Titusville (Pennsylvania, United States) in 1859, ensures that our empirical results do not suffer from any sample-selection bias.

As far as the econometric approach is concerned, we rely on a machine-learning approach, known as random forests (Breiman, 2001), which in turn has two main advantages. First, random forests can accurately analyze the links between changes in the UK unemployment rate and a large number of predictors in a full-fledged data-driven manner. Second, random forests automatically capture potential nonlinear links between the unemployment rate and its predictors, including uncertainties (Kandemir Kocaaslan, 2019; Christou et al., 2020; Kocaarslan et al., 2020), as well as any interaction effects between the predictors. To the best of our knowledge, this is the first paper to analyze the role of oil-price uncertainty, over and above a host of other macroeconomic and financial uncertainties, in forecasting change in the unemployment rate of the UK spanning over 150 years of monthly data using a machine-learning approach. It must be highlighted that most of the in-sample-based analyses of oil-price uncertainty have concentrated on movements in output, with only Rafiq et al. (2009), Kandemir Kocaaslan (2019) and Kocaarslan et al. (2020) depicting the negative impact of oil-price volatility on the unemployment rates of Thailand and the United States. While Rafiq et al. (2009) use realized volatility of oil returns in a vector autoregressive (VAR) model to capture a negative effect on the unemployment (and investment) of Thailand, Kandemir Kocaaslan (2019) rely on a generalized autoregressive conditional heteroskedasticity (GARCH)-in-mean VAR model of Elder and Serletis (2009) to show the same for the United States. Kocaarslan et al. (2020) rely on a nonlinear autoregressive distributed lag (NARDL) approach to point out that reduced oil-price uncertainty (as captured by the implied volatility, i.e., the crude-oil volatility index (OVX), and conditional volatility obtained from a GARCH model) leads to a decrease in unemployment, whereas an increase in oil-price uncertainty does not have an impact. While not related to oil uncertainty, Christou et al. (2020) analyze the effect of financial uncertainty, i.e., corporate bond spread and GARCH-based conditional stock market volatility, on UK's unemployment rate using a breaks-based VAR, and document a negative time-varying effect over the monthly historical period of 1855 to 2016.

At this stage, we must point out that Junttila and Vataja (2018), Aye et al. (2019b), and Salisu et al. (2022) have highlighted the important role played by general uncertainty in forecasting alternative measures of the performance of the aggregate real economy of the UK, while analyzing also other advanced economies within their data samples. Junttila and Vataja (2018) use measures of economic policy uncertainty (EPU) over the period from 1997 to 2016 to show that EPU improves the forecasting ability of linear predictive regressions for the growth of industrial production, especially for the period before the 2008 global financial crisis (GFC), but this metric of uncertainty is outperformed by financial market-related variables (i.e., slope of the yield curve and excess stock-market returns) during the GFC. This weak evidence of forecastability of industrial production growth of the UK due to EPU for the post-GFC period is also corroborated by Aye et al. (2019b) using a time-varying parameter-panel vector autoregressive (TVP-PVAR) model (involving twelve Organization for Economic Co-operation and Development (OECD) countries). Finally, Salisu et al. (2022) using data on gold-price volatility, as capturing uncertainty, given that gold is a well-established safe haven and higher trading in it during crises periods is likely to drive its returns and volatility, show that it forecasts annual real GDP growth over the period from 1311 to 2019, based on a linear predictive regression model. As can be seen, the existing studies on the UK have concentrated on only a specific measure of economic uncertainty while forecasting output growth, whereas we delve into the role of oil-market uncertainty for forecasting historical unemployment growth, by simultaneously including a wide-array of other variables, as well as their associated uncertainties, using a machine-learning approach that can accommodate many predictors in a data-driven, i.e., model-free manner, while also accounting for nonlinearities and interaction effects. In other words, our empirical research can be considered to be more robust compared to the existing studies on the nexus of uncertainty and predictions of macroeconomic variables.

We organize the remainder of our research as follows. In Section 2, we outline the basics of random forests. In Section 3, we briefly discuss our data. In Section 4, we present the results from our forecasting experiment. In Section 5, we conclude.

2. Random forests

Random forests are a machine-learning technique that makes it possible to inspect the predictive value of oil-price uncertainty for changes in the unemployment rate in the presence of several other predictors (including, e.g., stock-market and exchange-rate uncertainty). Moreover, random forests capture in a natural data-driven way potential interaction effects between these predictors. In addition, random forests

capture in a data-driven way any nonlinearities present in the data. Accounting for such nonlinearities is potentially important because economic theory suggests that the effect of uncertainty on economic variables is likely to be nonlinear (that is, the effect of high uncertainty is likely to differ from the effect of low or moderate uncertainty). In addition, an intuitive algorithm governs the computation of random forests, which clearly is an advantage given that random forests have not yet been extensively studied by the economics profession.

A random forest consists of an ensemble of individual regression trees. We, therefore, start with a description of how a regression tree, T, is computed. The core idea underlying the computation of a regression tree is that the branches of the tree partition the space of predictors, $\mathbf{x} = (x_1, x_2, \ldots)$, into l non-overlapping regions, R_l (for an in-depth introduction, see Hastie et al. (2009); we largely use their notation). The branches and the resulting regions are formed in a top-down way starting at the root of the tree and then moving to its leaves by means of a recursive search-and-split algorithm. The easiest way to describe how this algorithm works is to start at the root of a regression tree.

At the root of a tree, the search-and-split algorithm loops over the array of predictors. For every predictor, s, the algorithm then loops over its realizations. Every realization of a predictor is a candidate for a splitting point, p. For every combination of a predictor and a splitting point, the search-and-split algorithm forms two half-planes, $R_1(s,p) = \{x_s | x_s \leq p\}$ and $R_2(s,p) = \{x_s | x_s > p\}$. The algorithm identifies the optimal half-planes and, thereby, the optimal combination of a predictor and a splitting point, by minimizing the following squared-error loss criterion:

$$\min_{s,p} \left\{ \min_{\bar{f}_1} \sum_{x_s \in R_1(s,p)} (f_i - \bar{f}_1)^2 + \min_{\bar{f}_2} \sum_{x_s \in R_2(s,p)} (f_i - \bar{f}_2)^2 \right\}, \tag{1}$$

where the index i identifies those data of the predictand, f (in the context of our analysis, the change in the unemployment rate at some forecast horizon), that belong to a half-plane, and $\bar{f}_k = \text{mean}\{f_i | x_s \in R_k(s,p)\}, k = 1,2$ denotes the half-plane-specific mean of the predictand (for notational convenience, we do not use a time index).

The intuition behind Eq. (1) is straightforward: The outer minimization loops over all pairs of s and p, while the inner minimization identifies, for a given pair of s and p, the half-plane-specific means of the predictand by minimizing the half-plane-specific squared error loss. At this point, the simple regression tree basically resembles a conventional least-squares regression model with one predictor variable, where two dummy variables capture whether the predictor variable takes on a value above or below the splitting point.

When one repeatedly applies the search-and-split algorithm to grow a more complex regression tree, this amounts to replacing the two dummy variables with several additional dummy variables that partition the predictor space into finer and finer intervals. For example, the solution of the minimization problem given in Eq. (1) gives the first optimal splitting predictor, the first optimal splitting point, and the two region-specific means of the predictand. One then solves again the minimization problem given in Eq. (1), but now separately for the two optimal top-level half-planes, $R_1(s,p)$ and $R_2(s,p)$, resulting in up to two second-level optimal splitting predictors and their optimal splitting points, and four second-level region-specific means of the predictand. Further application of the search-and-split algorithm then renders it possible to form a regression tree whose hierarchical structure becomes increasingly complex.

Once the search-partition algorithm stops, the data on the predictors are sent from the root to the terminal nodes down a regression tree along its nodes and branches, and a forecast of the predictand can then be computed by means of the region-specific means at the terminal nodes of a regression tree. When the regression tree has L regions, a forecast is computed as follows ($\mathbf{1}$ denotes the indicator function):

$$T\left(\mathbf{x}_i, \{R_l\}_1^L\right) = \sum_{l=1}^{L} \bar{f}_l \mathbf{1}(\mathbf{x}_i \in R_l). \tag{2}$$

While repeated application of the search-and-split algorithm results in fine granular forecasts of the predictand, growing an increasingly complex regression tree does not always improve forecast accuracy. Again, the intuition is straightforward when one considers the analogy to a least-squares regression model that features several dummy variables to partition the predictor space into various subintervals. No empirical researcher would use such a complex model for forecasting purposes because it obviously suffers from a severe overfitting problem. In a similar vein, the increasing complexity of the hierarchical structure of a large regression tree naturally results in an overfitting and data-sensitivity problem.

Random forests provide a modeling platform to overcome this data-sensitivity problem. To this end, random forests combine a large number of (random) regression trees. A random forest is grown in two steps: (1) A large umber of bootstrap samples is generated from the data. Sampling is done with replacement. (2) A random regression tree is estimated on every bootstrap sample. A random regression tree selects for every splitting decision a randomly chosen subset of the predictors. This random sampling of predictors curbs the influence of influential predictors on tree growing. Growing a large number of random regression trees using this two-step recipe decorrelates the forecasts obtained from the individual (random) regression trees that form a random forest, while averaging the forecasts across the random regression trees stabilizes the forecasts computed by means of a random forest.

3. Data

Our raw data set covers the monthly period from 1859:09 to 2020:05, with the start date driven by the West Texas Intermediate (WTI) oil price, and the end date governed by the unemployment rate at the time of writing of this paper. Data on the unemployment rate, the consumer price index (CPI), the pound-dollar exchange rate, the short-term monetary policy rate, the long-term government bond yield (used to compute the term-spread as the difference between the long-term and short-term interest rates), and a dummy defining recession dates until 2016 are derived from the "A Millennium of Macroeconomic Data", which is maintained by the Bank of England,[2] and updated until the recent dates using the Main Economic Indicators (MEI) database of the Organization for Economic Co-operation and Development (OECD).[3] It must be pointed out that, in order to account for the zero lower bound (ZLB) situation and unconventional monetary policy decisions during the GFC, European sovereign debt crisis, and COVID-19 outbreak, we use the Shadow Short Rate (SSR) of Wu and Xia (2016),[4] derived by modeling the term structure of the yield curve, to capture the monetary policy stance during these episodes, instead of the Bank Rate. WTI oil price and the All Share Stock Index (ALSI) are derived from the Global Financial data.[5] The WTI data is in US dollars and is converted to British pound by dividing with the dollar-pound nominal exchange rate. To ensure stationarity, the unemployment rate and the short-term interest rate are first-differenced, while we use the first-difference of the natural logarithmic values of the CPI, the ALSI, the exchange rate, and the WTI oil price in local currency. Finally, following Sadorsky (1999), and the extant literature on oil-price uncertainty, as well as macroeconomic and financial uncertainty, we fit a Generalized Autoregressive Conditional Heteroskedasticity (GARCH(1,1)) model to obtain the conditional volatilities of the first-difference of the unemployment rate and interest rate, inflation rate (i.e., the growth

[2] The data is available at: https://www.bankofengland.co.uk/statistics/research-datasets.

[3] https://www.oecd.org/sdd/oecdmaineconomicindicatorsmei.htm.

[4] The data is available for download from: https://sites.google.com/view/jingcynthiawu/shadow-rates?authuser=0.

[5] http://www.globalfinancialdata.com/.

Table 1
Baseline forecasting statistics.

Panel A: RMSFE ratios				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	1.0023	1.0086	1.0230	1.0189
240	1.0104	1.0155	1.0028	1.0112
480	0.9903	0.9939	1.0210	1.0231
Panel B: MAFE ratios				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	0.9945	1.0126	1.0108	1.0181
240	1.0110	1.0085	1.0020	1.0187
480	0.9920	0.9931	1.0132	1.0128

Note: This table reports results of RMSFE (MAFE) ratio obtained by dividing the RMSFE (MAFE) of a model that does not feature oil-price uncertainty in the array of predictors by the RMSFE (MAFE) of a model that can use oil-price uncertainty to build random forests. The parameter h denotes the forecast horizon (in months).

Table 2
Diebold–Mariano test results.

Panel A: L1 loss				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	0.7649	0.0720	0.1090	0.0062
240	0.0591	0.1343	0.3787	0.0427
480	0.8248	0.8486	0.0305	0.1251
Panel B: L2 loss				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	0.4471	0.1279	0.0509	0.0009
240	0.1253	0.0923	0.4134	0.1055
480	0.9183	0.8562	0.0395	0.0989

Note: This table reports results (p-values) of Diebold–Mariano tests under L1 (loss depends on the absolute forecast error) and L2 (loss depends on the squared forecast error) loss. The null hypothesis is that the accuracy of forecasts extracted from of a model that does not feature oil-price uncertainty in the array of predictors is equal to the accuracy of forecast computed by means a model that can use oil-price uncertainty to build random forests. The alternative hypothesis is that the latter forecasts are more accurate than the former. The parameter h denotes the forecast horizon (in months).

Table 3
Clark–West test results.

Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	0.1866	0.0033	0.0221	0.0001
240	0.0282	0.0213	0.1109	0.0059
480	0.4444	0.3964	0.0260	0.0348

Note: This table reports results (p-values) of Clark–West tests loss. t-statistics of the Clark–West tests are based on Newey–West standard errors. The parameter h denotes the forecast horizon (in months). The parameter h denotes the forecast horizon (in months).

rate of the CPI), and (log-returns of) the ALSI, exchange rate and WTI oil price. Given the data transformation undertaken for the sake of mean-reversion, our effective sample covers the period 1859:10 to 2020:05. The dependent variable is the change in the unemployment rate, which we aim to forecast, while the (13) predictors are the recession dummy, inflation rate, change in the monetary policy rate, term-spread, stock returns, oil returns, exchange-rate returns, and the conditional volatilities of changes in the unemployment rate and the interest rate, inflation rate, oil, stock, and exchange-rate returns.

4. Empirical results

Given the length of our sample period, we account for a potentially time-varying predictive value of oil-price uncertainty for changes in the UK unemployment rate by estimating random forests on rolling-estimation windows of length 120, 240, and 480 months. We forecast changes in the UK unemployment rate at four forecast horizons (h): 1 month, 3 months, 6 months, and 12 months, where we forecast the sum of the change in the unemployment rate (that is, the accumulated changes) when we study the latter three forecast horizons, and we construct the data matrix such that the number of forecasts is the same for all four forecast horizons. We use the R language for statistical computing (R Core Team, 2019) to carry out our forecasting experiments, where we estimate random forests by means of the add-on package "grf" (Tibshirani et al., 2020). When we move the rolling-estimation windows across our data, we use cross validation to optimize the number of predictors randomly selected for splitting, the minimum node size of a tree, and the parameter that governs the maximum imbalance of a node.[6] We use 2,000 random regression trees to grow a random forest.

In order to set the stage for our analysis, we report baseline forecasting statistics in Table 1. Specifically, we report in Panel A the ratio of the root-mean-squared forecasting error (RMSFE) of a model that neglects oil-price uncertainty and the RMSFE of a model that includes oil-price uncertainty in the array of predictors. A RMSFE ratio that exceeds unity, thus, indicates that oil-price uncertainty improves forecast accuracy under squared-error (L2) loss. In Panel B, we replace the L2 loss function with an L1 loss function and report the ratio of the mean-absolute forecast errors (MAFE) implied by the models with/without oil-price uncertainty in the array of predictors. Again, a value of the MAFE ratio above unity shows that oil-price uncertainty helps to improve forecast accuracy.

While the predictive value of oil-price uncertainty at the two shorter forecast horizons (that is, $h = 1, 3$) depends on the length of the rolling-estimation window, the RMSFE and MAFE ratios are consistently larger than unity for the two longer forecast horizons ($h = 6, 12$). In fact, the RMSFE and MAFE ratios attain their maximum in four out of the six considered cases for $h = 12$. Hence, the RMSFE and MAFE ratios provide evidence that oil-price uncertainty has predictive value for subsequent changes in the unemployment rate, especially when we consider the longer forecast horizons.

The results of Diebold and Mariano (1995) tests corroborate the predictive value of oil-price uncertainty at the longer forecast horizon. We report the test results (p-values) under the L1 and L2 loss functions for the well-known modified Diebold–Mariano test proposed by Harvey et al. (1997) in Table 2.[7] The significant test results (assuming a marginal significance level of 10%) mainly can be observed for the two long forecast horizons, $h = 6$ and $h = 12$.

In Table 3, we summarize the results for the test proposed by Clark and West (2007). Four out of six test results are significant for the two longer forecast horizons, where we use robust standard errors to assess the significance of the test results. In addition, we find three significant test results for the two shorter forecast horizons. These three significant test results have significant counterparts in Table 2, albeit the level of significance is weaker for the Diebold–Mariano test than for the Clark–West test and depends on the assumed loss function. Fig. 1 plots the relative importance of both the oil price and oil-price uncertainty over time, where we focus on the long rolling-estimation window (480 months) and the long forecast horizon ($h = 12$). Relative importance is defined as the weighted sum of how often a predictor is used for splitting the predictor space at each partitioning level in a random forest. Oil-price uncertainty apparently was relatively more important than changes in the oil price itself for most of the sample period. Moreover, our results show that, from a historical perspective,

[6] The "grf" package also allows different subsamples to be used for constructing a tree and for making predictions. We deactivate this option, as in a classic random forest.

[7] We report p-values computed using the R package "forecast" (Hyndman and Khandakar, 2008; Hyndman, 2017).

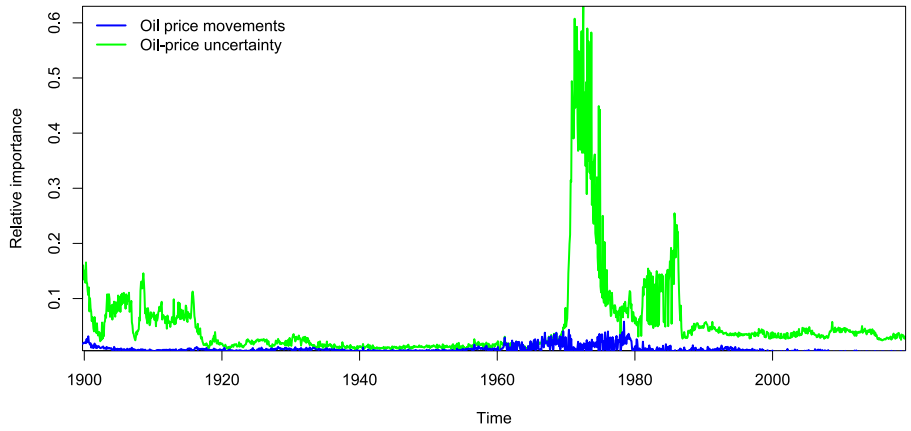


Fig. 1. Relative importance of oil-related predictors. Window length 480 months. Predictor importance is computed for $h = 12$. Predictor importance is defined as the weighted sum of how often a predictor is used for splitting the predictor space at each partitioning level in a random forest.

oil-price uncertainty was relatively important before 1900 and then again during two periods of time after the World War II. The first roughly covers the 1970s while the second period covers the 1980s, that is, in the periods of time during which the first and second oil-price shocks hit Western industrialized countries.

As a final extension, we compare the forecast performance of random forests, a nonlinear machine-learning estimator, with the forecasting performance of the Lasso estimator, a popular linear machine-learning estimator. The loss function motivating the Lasso (least absolute shrinkage and selection operator), proposed by Tibshirani (1996), consists of two parts. As for the first part, coefficients are to be chosen so as to minimize the squared distance between a linear combination of the predictors and the predictand, as in the case of the ordinary least squared (OLS) technique. The second part is the penalty part, which consists of the sum of the absolute magnitudes of the coefficients. The second part is added to the first part and, thereby, facilitates the search for a parsimonious forecasting model in a data-rich modeling environment. We use the R add-on package "glmnet" (Friedman and Tibshirani, 2010) to compute Lasso forecasts.

In Table 4, we report in Panel A (Panel B) the RMSFE (MAFE) ratios that we obtain when we divide the RMSFE (MAFE) of a model that does not feature oil-price uncertainty in the array of predictors by the RMSFE (MAFE) of a model that can use oil-price uncertainty to obtain the Lasso estimator. Given that the reported ratios exceed unity in the majority of cases, we conclude that adding oil-price uncertainty to the array of predictors helps to improve the forecasting performance of the linear Lasso estimator, especially at the two longer forecast horizons of six and twelve months. In Panel C (Panel D), we report the RMSFE (MAFE) ratio that we obtain when we divide the RMSFE (MAFE) of the Lasso model by the RMSFE (MAFE) of the random-forest model. Both models can use oil-price uncertainty in the array of predictors and, thus, we use Panels C and D to compare the forecasting performance of the linear Lasso estimator with that of the nonlinear random forests. The message to take home is that random forests, our preferred method in this research, outperform the Lasso estimator in terms of both the RMSFE and the MAFE statistics. Crucially, we observe that this improvement in forecasting performance is particularly strong for the longer forecast horizons, which, in turn, are those forecast horizons for which the evidence that oil-price uncertainty does have predictive value is relatively strong.

To sum up, the results of both the basic RMSFE and MAFE statistics and the formal statistical tests show that oil-price uncertainty has predictive value for changes in the UK unemployment rate, especially at the two longer (six and twelve months) forecast horizons. Moreover, the relative importance of oil-price uncertainty as a predictor changed over time. Focusing on the long forecast horizon, for which evidence of predictability is strong, and a long rolling-estimation window, such that plenty of data can be used to fit the random forests and to recover potential nonlinearities in the data and interactions among the predictors, we find evidence that relative importance changed over time, that relative importance was high following the oil-price shocks of the 1970s and 1980s, and that it tended to be higher than the relative importance of changes in the oil price during most of the sample period. Moreover, we also find evidence of predictive value of oil-price uncertainty for changes in the UK unemployment rate, again particularly at the longer forecast horizons, when we study the Lasso estimator, which is a popular alternative to random forests. Random forests though produce better results than the Lasso estimator in terms of the popular RMSFE and MAFE statistics, and the dominance of random forests tends to strengthen as the forecast horizon increases. Hence, nonlinearities in the data and interaction effects among the

Table 4
Forecasting statistics for the Lasso estimator.

Panel A: RMSFE ratios (relative to baseline)				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	1.0023	1.0086	1.0230	1.0189
240	1.0104	1.0155	1.0028	1.0112
480	0.9903	0.9939	1.0210	1.0231
Panel B: MAFE ratios (relative to baseline)				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	0.9945	1.0126	1.0108	1.0181
240	1.0110	1.0085	1.0020	1.0187
480	0.9920	0.9931	1.0132	1.0128
Panel C: RMSFE ratios (relative to random forest)				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	1.0602	1.2365	1.6507	1.4267
240	1.1369	1.2190	1.3343	1.2828
480	1.1008	1.3061	1.3928	1.4149
Panel D: MAFE ratios (relative to random forest)				
Window	$h = 1$	$h = 3$	$h = 6$	$h = 12$
120	1.0297	1.1703	1.2604	1.2865
240	1.1003	1.1952	1.2587	1.3311
480	1.1916	1.3648	1.4027	1.4465

Note: This table reports in Panel A (Panel B) results of RMSFE (MAFE) ratio obtained by dividing the RMSFE (MAFE) of a baseline model that does not feature oil-price uncertainty in the array of predictors by the RMSFE (MAFE) of a model that can use oil-price uncertainty to obtain Lasso forecasts. Panel C (Panel D) summarizes the RMSFE (MAFE) ratio obtained by dividing the RMSFE (MAFE) of the Lasso model by the RMSFE (MAFE) of the random-forest model, where both models can use oil-price uncertainty in the array of predictors. The parameter h denotes the forecast horizon (in months).

predictors appear to be crucial features of the data that researchers should take into account when empirically studying the link between oil-price uncertainty and subsequent changes in the UK unemployment rate.

5. Conclusion

We have used a long-span macroeconomic and financial data set and random forests to shed light on the predictive value of oil-price uncertainty for changes in the UK unemployment rate at various forecast horizons. Random forests have several advantages: They are a data-driven approach, they account in a natural way for a potential nonlinear predictive link between oil-price uncertainty and changes in the unemployment rate, and the make it possible to control, in a unified framework, for the impact of several other macroeconomic and financial data. Our results show that oil-price uncertainty does have predictive value for movements in the unemployment rate, especially at the longer forecast horizons of six and twelve months. We also have documented the time-varying relative importance of oil-price uncertainty for changes in the unemployment rate.

Our results have important implications for academics, investors and policymakers. For academics, our findings indicate that, in general, machine-learning approaches provide a viable tool for conducting forecasting experiments involving large number of predictors. More specifically in our specific-context, researchers must also realize that machine-learning methods that control for nonlinearities are preferable over linear versions of the same in forecasting historical unemployment rates of the UK. Since the future path of the unemployment rate, and hence employment and output, is a major determinant of investment decisions (Rapach and Wohar, 2007), our results can be used by investors to make decision on the "optimal" level of capital input that should be purchases for the production process to maximize profit. Finally, from a policy perspective, our empirical results highlight that policymakers in general, and central banks in particular, need to monitor oil-price uncertainty, over and above macroeconomic and financial uncertainties, when forecasting developments in the real economy while making their policy decisions. More specifically, given that oil-price uncertainty carries additional information over and above other macroeconomic and financial predictors, as well as their respective uncertainties, for the future path of the unemployment rate, policymakers can incorporate the variability of oil in their setting of interest rates to ensure that unemployment rate can be controlled appropriately to prevent possible recessions.

In terms of future research, it would be interesting to apply the method we have used in this research to study uncertainty of oil and other commodity prices, and their impact on other developed and emerging market economies, but of course this would not involve the entire history of the petroleum industry, but a much shorter post World War II sample of data available for these economies and commodity markets.

CRediT authorship contribution statement

Rangan Gupta: Conceptualization, Data curation, Methodology, Writing – original draft, Writing – review & editing. **Christian Pierdzioch:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Afees A. Salisu:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

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